Upstream Kernel CI

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Landscape: projects

kernelci.org

Red Hat’s Continuous Kernel Integration

Linaro Kernel Functional Testing

01 INTEL OPEN SOURCE .org

0-Day - Linux Kernel Performance

syzkaller

Landscape: Tests

- Linux kernel test tools:
  - Kselftest, coccinelle, KASAN, UBSAN, KUnit...
- Comprehensive test suites:
  - Linux Test Project
- Subsystem test suites
  - v4l2-compliance, i-g-t, xfstests...

Commonalities

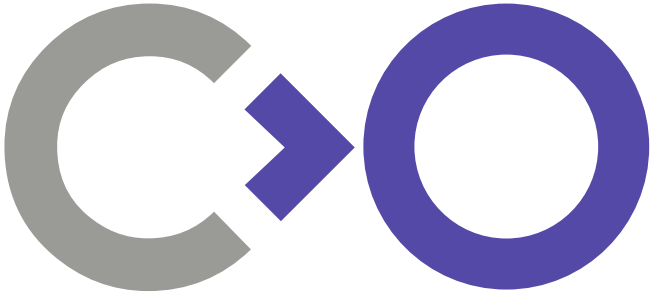
- Monitor git branches or patches
- Build kernels
- Run tests
- Process the results
- Send emails
- Report on a dashboard

So many wheels

- Each project has some good tools and features
- Combining them could lead to more possibilities
- Different purposes:
  - CPU architecture (Intel...)
  - Boards (96Boards...)
  - Distribution (Fedora...)
  - Kernel frameworks (Kunit...)

Ideas

- Common test results database
- Common set of tools to orchestrate CI
- Test coverage “map”
- Shared hardware pools
- Public APIs to let components talk to each other
  - See also: “Open Testing Philosophy”



Let’s talk!

Photo credits

- 3. test equipment: https://www.flickr.com/photos/129143611@N03/16348438725/
- 4. grinder: https://www.flickr.com/photos/16041363@N00/2338931226/
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